Assignment Name: Project 1 Part 3

Class Name and Section: ECE452-002 **Date Submitted:** 03/15/2026

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Introduction & Procedure:

In Part 3, we analyzed the impact of different activation functions on the classification accuracy of the MLP trained in Part 2. Using the same network architecture and training configuration from Part 2 (256 → 128 → 64 → 10, 100 epochs, MNIST dataset resized to 16x16), we retrained the network three times, each time using a different hidden layer activation function: tansig, logsig, and poslin (ReLU). All other hyperparameters were kept identical to isolate the effect of the activation function.

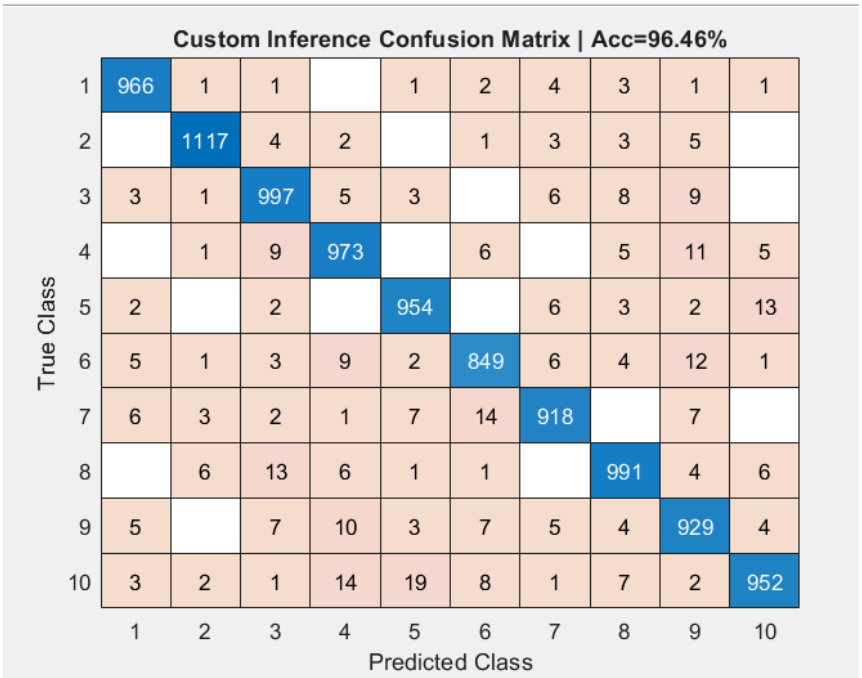
Results and Discussion:

Based on our experiments, we found that tansig performed the best at 96.46%, followed closely by logsig at 96.24%. Poslin, or ReLU, performed significantly worse at 70.51%, which is a dramatic drop compared to the other two.

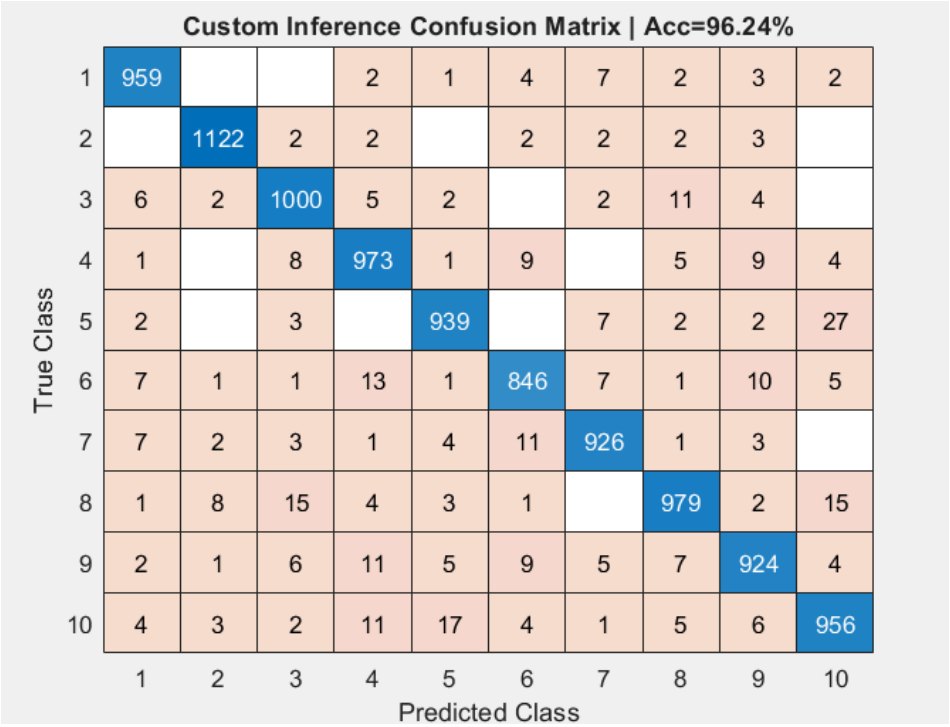
Tansig and logsig both performed similarly because they are both smooth, bounded activation functions that produce well-behaved gradients throughout training. Tansig outputs values in the range [-1, 1] and is zero-centered, which generally leads to slightly faster and more stable convergence compared to logsig, which outputs values in [0, 1]. This explains the small but consistent accuracy gap between the two.

Poslin (ReLU) performed significantly worse for this specific setup. Looking at the confusion matrices, digits 1 and 8 received 0% accuracy, meaning the network completely failed to classify those two classes. This is a classic symptom of the dying neuron problem, where ReLU neurons get stuck outputting zero for all inputs and stop contributing to learning. This is more likely to happen with MATLAB's scaled conjugate gradient optimizer compared to Adam, which has adaptive learning rates that can help recover dying neurons. The combination of poslin with this optimizer and initialization led to a partially collapsed network that could not recover within 100 epochs.

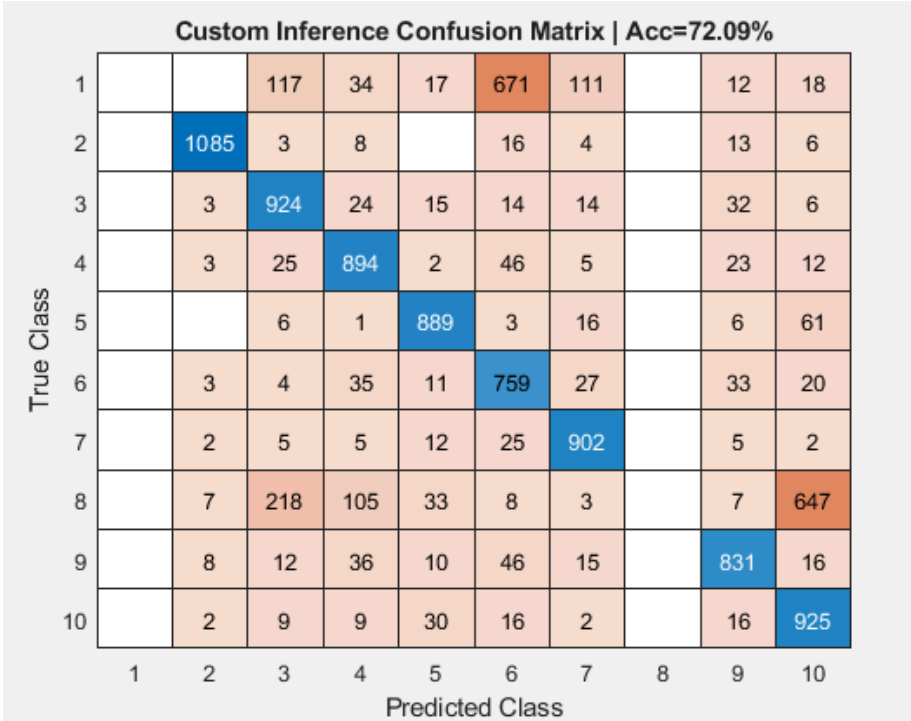
Activation	Toolbox Accuracy	Custom Inference Accuracy
tansig	96.76%	96.46%
logsig	96.24%	96.24%
poslin	72.09%	72.09%



Tansig (tanh) Confusion Inference Matrix



Logsig Confusion Inference Matrix



Poslin (ReLU) Confusion Inference Matrix

Conclusion:

In conclusion, our experiments confirmed that activation function selection has a measurable effect on the performance of a customized MLP. Based on our results, we found that the Tanh activation was the most effective choice for this project due to its having the highest test accuracy, as well as the most reliable classification behavior. This was in contrast to the network structure we used in part 1, which was mainly ReLU activation throughout the hidden layers, and then a Softmax (which we later learned was embedded in the Cross-Entropy Loss function). In this part, we attempted to train a full network model using the ReLU activation instead of Tanh, and found that it performed quite badly, despite its success in previous parts. This is especially clear in the confusion matrix above, where the network (which we believe had *dying neurons*) failed to learn any significant features about the numbers 1 and 8 in the 100 training epochs.

Everything considered, we believe that this project was very beneficial, helping us understand the structure of Neural Networks and also giving us hands-on experience with the MATLAB Deep Learning Toolkit, as well as various deep learning libraries for Python.

References:

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